0.状态方程/动理论 $PV=nRT=NkT$ $P=\frac{1}{3}nm\overline{v^2} \Rightarrow K_{平动}=\frac{3}{2}kT$, $v_{rms}=\sqrt{\overline{v^2}}=\sqrt{3kT/m}\approx 500\,m\cdot s^{-1}$ (air). 能均分 $U=N\cdot f\cdot\frac{1}{2}kT$ 单原子 $f=3$, 双 $f=5$, 晶体 $f=6$

I.热一 $\Delta U=Q+W$ 准静态压缩功: $dW=-PdV$, $W=-\int_{V_i}^{V_f}P(V)dV \Rightarrow$ 等T, $W=NkT\ln V_i/V_f$; 绝热: $dU=-PdV=\frac{1}{2}fNkdT \Rightarrow \frac{1}{2}f\ln T_f/T_i=-\ln V_f/V_i \Rightarrow VT^{f/2}=const.$

$PV^\gamma=const.$ $\gamma=f+2/f$. $d(PV)=NkdT \Rightarrow dT/dP=2/f+2\cdot T/P$ 热容: $C_V=(\partial U/\partial T)_V=\frac{1}{2}Nfk$ 若能均分 $C_P=(\partial H/\partial T)_P=(\partial U/\partial T)_P+P(\partial V/\partial T)_P=C_V+Nk$

Fourier 导热: $dQ/dt=-k_tA\,dT/dx$, $J_q=-k_t\,dT/dx \Rightarrow k_t=\frac{1}{4}\frac{fP}{T}\cdot\bar v\cdot l\propto\sqrt{T}$ Fick 扩散: $J=-D\,dn/dx$, $D\sim l^2/2\Delta t$ Newton 黏度: $\sigma=|F_x|/A=\eta\cdot du_x/dz$, $\eta=\frac{1}{2}nml|\bar v|\propto\sqrt{T}$

II.热二 Stirling 近似: $N!\approx N^Ne^{-N}\sqrt{2\pi N}$, $\ln N!\approx N\ln N-N$ 二态顺磁: $\Omega(N_\uparrow)=N!/N_\uparrow!(N-N_\uparrow)!$ Einstein 固体: N 振子, q 个能量量子. $\Omega(N,q)=(q+N-1)!/q!(N-1)!$

$q\gg N$ (高T), $\Omega(N,q)\approx (q+N)!/q!N! \Rightarrow \ln\Omega\approx(q+N)[\ln q+\ln(1+N/q)]-q\ln q-N\ln N\approx N\ln q/N+N \Rightarrow \Omega\approx(eq/N)^N$ $q\ll T$ 时 $\Omega\approx(eN/q)^q$. 两个相作用固体, $\Omega=(e/N)^{2N}q_A^Nq_B^N$

$\Rightarrow \Omega_{max}=(e/N)^{2N}(q/2)^{2N}$. 峰周围 $\Omega=(e/N)^{2N}(q^2/4-x^2)^N\approx\Omega_{max}e^{-4x^2/q^2} \Rightarrow$ 宽 $q/\sqrt{N}$ 理想气体: $\Omega_N=\frac{1}{N!}\frac{V^N}{h^{3N}}\times V_p$ 的 3N 维超球面积 $=\frac{1}{N!}\frac{V^N}{h^{3N}}\times\frac{2\pi^{3N/2}}{(3N/2-1)!}(2mU)^{\frac{3N-1}{2}}$

$\approx\frac{1}{N!}\frac{V^N}{h^{3N}}\cdot\frac{\pi^{3N/2}}{(3N/2)!}(2mU)^{3N/2}\propto V^NU^{3N/2} \Rightarrow S=k\ln\Omega\approx Nk\{\ln[\frac{V}{N}(\frac{4\pi mU}{3Nh^2})^{\frac{3}{2}}]+\frac{5}{2}\}$. $\Omega=f(N)\cdot V^N\cdot U^{\frac{3}{2}N} \Rightarrow V_i\to V_f$, $\Delta S=Nk\ln V_f/V_i$ 混合熵: $N(1-x)A+NxB$

$\Delta S=-Nk[x\ln x+(1-x)\ln(1-x)]$

III.相互作用 (1)温度 $1/T=(\partial S/\partial U)_{N,V}$ Einstein 固体 $S=Nk[\ln q/N+1]=Nk\ln U-Nk\ln(N\varepsilon)+Nk \Rightarrow T=(\partial S/\partial U)^{-1}=U/Nk \Rightarrow U=\varepsilon q=NkT$

(2) 热三. $dS=dU/T=Q/T\,(\Delta V=0)=C_VdT/T \Rightarrow \Delta S=S_f-S_i=\int_{T_i}^{T_f}C_V/T\,dT \Rightarrow S=S(0)+\int_0^{T_f}C_V/T\,dT$. $T=0$ 处 $C_V=0$ (3) 顺磁: N 个独立偶极子处于 $+z$ 的 $\vec B$

$\downarrow+\mu B$, $\uparrow-\mu B \Rightarrow U=\mu B(N-2N_\uparrow)$, $M=\mu(N_\uparrow-N_\downarrow)=-U/B$, $\Omega(N_\uparrow)=N!/N_\uparrow!N_\downarrow!$ 这里会有 $T=(\partial S/\partial U)^{-1}<0$ (当总能量有限, $T=\infty$ 是随机性最高) $x=\mu B/kT$

求解. $T^{-1}=(\partial S/\partial U)=\partial N_\uparrow/\partial U\cdot\partial S/\partial N_\uparrow=k/2\mu B\ln(N-U/\mu B)/(N+U/\mu B) \Rightarrow U=-N\mu B\tanh x$, $M=N\mu\tanh x$, $C_B=(\partial U/\partial T)_{N,B}=Nk\,x^2/\cosh^2(x)$, $S=Nk[\ln(2\cosh x)-x\tanh x]$

现实顺磁体: $\mu B/kT\ll1 \Rightarrow M\approx N\mu^2B/kT$ (4) 力/扩散平衡 $P=T(\partial S/\partial V)_{U,N}$, $\mu=-T(\partial S/\partial N)_{U,V} \Rightarrow dS=dU/T+PdV/T-\mu dN/T \Rightarrow dU=TdS-PdV+\mu dN$

$\Rightarrow$ 单原子分子理想气体: $\mu=-kT\ln[V/N(2\pi mkT/h^2)^{3/2}]$ (5) Gibbs-Duhem 关系: 广延量 $U[\lambda S,\lambda V,\lambda N]=\lambda U[S,V,N]$, 对 λ 求导并令 $\lambda=1$.

$\partial U/\partial S\,S+\partial U/\partial V\,V+\partial U/\partial N\,N=U \Rightarrow TS-PV+N\mu=U \Rightarrow TdS+SdT-PdV-VdP+\mu dN+Nd\mu=dU \Rightarrow SdT-VdP+Nd\mu=0$ $(\Rightarrow V(\partial P/\partial T)_\mu=S,\ V(\partial P/\partial\mu)_T=N)$

IV.热机. (1)热机 $T_h\xrightarrow{Q_h}$ 热机 $\xrightarrow{W,\ Q_c}T_c$. 效率 $e=W/Q_h=1-Q_c/Q_h$. 热二要求 $\Delta S=-Q_h/T_h+Q_c/T_c\geq0 \Rightarrow Q_c/Q_h\geq T_c/T_h \Rightarrow e\leq1-T_c/T_h$.

(2)制冷机 $T_h\xleftarrow{Q_h}$ 制冷机 $\xleftarrow{W,\ Q_c}T_c$. 性能系数 $COP=Q_c/W=1/(Q_h/Q_c-1)$. 同样 $Q_h/Q_c\geq T_h/T_c \Rightarrow COP\leq T_c/(T_h-T_c)$ (3) 真实热机. 可逆热机效率最高.

e.g. Carnot (等温交换热, 绝热可逆变温, 不产生多余S) Otto 机. $e=1-\frac{T_1-T_4}{T_2-T_3}$ Diesel 机 $e=1-\frac{\alpha^\gamma-1}{\gamma r^{\gamma-1}(\alpha-1)}$, $r=V_1/V_2$, $\alpha=V_3/V_2$

V.自由能与化学热力学 (1) $H\equiv P+UV$ 凭空创造一个系统并放置在环境中所需的总能 $F\equiv U-TS$ 凭空创造一个系统所需要提供的能量（环境会提供一部分

$G\equiv U-TS+PV$ 凭空创造一个系统，并放置在环境中所需要提供的能量 恒T变化 $\Delta F\leq W$. 再固定P, $\Delta G\leq W_{非}$. 孤立系统 S 总↗, 非孤立系统, $dS_{tot}=dS+dS_R$

$dS_R=dU/T+PdV/T-\mu dN/T$. 若 V,N 不变, $dS_{tot}=dS-dU/T=-dF/T \Rightarrow$ 最小化 F; 若 P,N 不变, $dS_{tot}=dS-dU/T-PdV/T=-dG/T \Rightarrow$ 最小化 G.

(2) 热力学恒等式 $dU=TdS-PdV+\mu dN$, $dH=TdS+VdP+\mu dN$, $dF=-SdT-PdV+\mu dN$, $dG=-SdT+VdP+\mu dN$ $\Rightarrow S=-(\partial F/\partial T)_{V,N}=-(\partial G/\partial T)_{P,N}$

$\mu=(\partial F/\partial N)_{T,V}=(\partial G/\partial N)_{T,P}$ etc. T,P 强度量, G 广延量 $\Rightarrow G\propto N \Rightarrow G=\mu N \Rightarrow \partial\mu/\partial P=N^{-1}\partial G/\partial P=V/N=kT/P \Rightarrow \mu(T,P)=\mu^\circ(T)+kT\ln P/P^\circ$

(3) Maxwell 关系. 对 $Z(x,y)$ $dz=Mdx+Ndy$ 是全微分 $\Leftrightarrow (\partial M/\partial y)_x=(\partial N/\partial x)_y \Rightarrow$ e.g. $dU=TdS-PdV+\mu dN \Rightarrow (\partial T/\partial V)_{S,N}=-(\partial P/\partial S)_{V,N}$

(4) Clausius-Clapeyron 关系. l-g 的 P-T 边界. $dG_l=dG_g \Rightarrow dP/dT=\frac{S_g-S_l}{V_g-V_l}=L/T\Delta V$. 若 L 不变, 忽略 V_l. $dP/dT\approx L/TV_g=LP/RT^2 \Rightarrow dP/P=L/R\,dT/T^2 \Rightarrow \ln P=-L/RT+C$

(5) van der Waals 模型 $(P+aN^2/V^2)(V-Nb)=NkT$. 固定 T,N, $(\partial G/\partial V)_{N,T}=V(\partial P/\partial V)_{N,T}=-NkTV/(V-Nb)^2+2aN^2/V^2 \Rightarrow G/NkT=-\ln(V-Nb)+\frac{Nb}{V-Nb}-\frac{2a}{kbT}\cdot\frac{Nb}{V}+C$

联立 $P/kT/b=\frac{Nb}{V-Nb}-\frac{a}{bT}(Nb/V)^2$, G-P 图出现 形状, 表明跳变 ⇒ l-g 相变, P-V 图 . 高温下 P-V 图单调 ⇒ 无相变 ⇒ 临界点 (T_c,P_c,V_c)

这点 $\partial P/\partial V=\frac{NkT}{(V-Nb)^2}+\frac{2aN^2}{V^3}=0$, $\partial^2P/\partial V^2=\frac{2NkT}{(V-Nb)^3}-\frac{6aN^2}{V^4}=0 \Rightarrow V_c=3Nb$, $kT_c=8a/27b$, $P_c=a/27b^2$. 压缩系数 $=P_cV_c/NkT_c=\frac{3}{8}$

(6) 混合物相变. $(1-x)A+xB$: $\Delta S_{mix}=-R[x\ln x+(1-x)\ln(1-x)] \Rightarrow$ 混合后 $G_{mix}=G-T\Delta S_{mix}$ (理想, 不改变 U,V) 此时 G-x 下凸, 故不分相. 若非理想, 混合 U↑

则 G-x 可能上凸 ⇒ 分相 (7) 稀溶液 纯A: $G=N_A\cdot\mu_0(T,P)$. 加入1个B, T,P 不变, $dG=dU+PdV-TdS=f(T,P)-kT\ln N_A$ (N_A 只和 dS 相关) ⇒ 加入 N_B 个不可分

$G=N_A\mu_0(T,P)+N_Bf(T,P)-N_BkT\ln N_A+(N_BkT\ln N_B-N_BkT) \Rightarrow \mu_A=(\partial G/\partial\mu_A)=\mu_0(T,P)-N_BkT/N_A$; $\mu_B=f(T,P)+kT\ln N_B/N_A$. 记 $m\equiv n_B/m_A$.

$\mu_B=\mu^\circ(T,P)+kT\ln m_B$ 渗透压: $\mu_0(T,P_1)=\mu_0(T,P_2)-N_BkT/N_A$, 近似 $\mu_0(T,P_2)\approx\mu_0(T,P_1)+(P_2-P_1)\partial\mu_0/\partial P \Rightarrow N_BkT/N_A=(P_2-P_1)\partial\mu_0/\partial P=(P_2-P_1)V/N_A$

$\Rightarrow P_2-P_1=N_BkT/V$. 蒸气压: 固定 T, $\mu_0(T,P)-N_BkT/N_A=\mu_g(T,P)$ 在蒸气压 P_0 处展开: $\mu_0(T,P_0)+(P-P_0)\partial\mu_0/\partial P-N_BkT/N_A=\mu_g(T,P_0)+(P-P_0)\partial\mu_g/\partial P \Rightarrow$

$(P-P_0)(V/N)_l-N_BkT/N_A=(P-P_0)(V/N)_g$, $(V/N)_l\approx0$, $(V/N)_g\approx kT/P_0 \Rightarrow P-P_0=-N_B/N_A\cdot P_0 \Rightarrow P/P_0=1-N_B/N_A$. 熔沸点: P 不变, $\mu_0(T,P)-N_BkT/N_A=\mu_g(T,P)$ 在纯A

沸点 T_0 处展开: $(T-T_0)\partial\mu_0/\partial T-N_BkT/N_A=(T-T_0)\partial\mu_g/\partial T \Rightarrow (T-T_0)(S/N)_l-N_BkT/N_A=(T-T_0)(S/N)_g$. $N\approx N_A$. $\Delta S=L/T_0 \Rightarrow T-T_0=N_BkTT_0/L\approx N_BkT_0^2/L$

同理熔点. $T-T_0=-N_BkT_0^2/L$ (8) 平衡: $e^{-\Delta G^\circ/RT}=K$, $\partial(\ln K)/\partial T=\Delta H^\circ/RT^2 \Rightarrow \ln K_2/K_1=\Delta H^\circ/R(1/T_1-1/T_2)$, 若 ΔH° 不变. 溶解: $m/P/P^\circ=e^{-\Delta G^\circ/RT}$

VI. Boltzmann 统计 (1) Boltzmann 因子. 可交换能量的系统和热库处于热平衡时, 某个特定微观态的概率: $P(s)\propto e^{-\beta E(s)}$, $\beta=(kT)^{-1}$, $Z=\sum_se^{-\beta E(s)}$, $P(s)=Z^{-1}e^{-\beta E(s)}$

(2) 平均值与涨落 $\langle E\rangle=Z^{-1}\sum_sE(s)e^{-\beta E(s)}=-Z^{-1}\partial Z/\partial\beta=-\frac{\partial}{\partial\beta}\ln Z$. 若有一系列粒子, $U=N\bar E$, 涨落 $\sigma_E^2=\langle E^2\rangle-\langle E\rangle^2=Z^{-1}\partial^2Z/\partial\beta^2-Z^{-2}(\partial Z/\partial\beta)^2=\frac{\partial}{\partial\beta}[Z^{-1}\partial Z/\partial\beta]=-\partial\langle E\rangle/\partial\beta$

$=-\partial\langle E\rangle/\partial T\cdot\partial T/\partial\beta=kT^2C \Rightarrow \sigma_E=kT\sqrt{C/k}$. $\langle E\rangle\propto N$, $C\propto N \Rightarrow \sigma_E/E\propto N^{-\frac{1}{2}}\to0\ (N\to\infty) \Rightarrow$ 涨落可忽略 (3) 双状态顺磁体: $Z=e^{\beta\mu B}+e^{-\beta\mu B}=2\cosh(\beta\mu B)$

$E=-Z^{-1}\partial Z/\partial\beta=-\mu B\tanh(\beta\mu B) \Rightarrow U=-N\mu B\tanh(\beta\mu B)$, $\langle\mu_z\rangle=\mu\tanh(\beta\mu B)$, $M=N\langle\mu_z\rangle=N\mu\tanh(\beta\mu B)$ (4) Einstein 固体: N 个振子, 允许能量为

$0, hf, 2hf\cdots \Rightarrow Z=\sum_{i=0}^\infty e^{-\beta\cdot ihf}=(1-e^{-\beta hf})^{-1} \Rightarrow \langle E\rangle=hf(e^{\beta hf}-1)^{-1} \Rightarrow U=Nhf(e^{\beta hf}-1)^{-1}$ (5) 双原子分子转动. 能级 $E=j(j+1)\varepsilon$, 有 $2j+1$ 个简并态.

$Z_{rot}=\sum_{j=0}^\infty(2j+1)e^{-j(j+1)\varepsilon\beta}\approx\int_0^\infty(2j+1)e^{-j(j+1)\varepsilon\beta}dj=(\varepsilon\beta)^{-1}=kT/\varepsilon \Rightarrow \langle E_{rot}\rangle=\beta^{-1}=kT$ (对应能均分 $f=2$) 若两个原子相同, 态数目 / $Z_{rot}\times\frac{1}{2}$, $\langle E_{rot}\rangle$ 不变

(6) 能均分. 适用于二次系统 $E(q)=cq^2$. 设离散, 间隔 Δq. $Z=\sum_qe^{-\beta E(q)}=\frac{1}{\Delta q}\sum_qe^{-\beta cq^2}\Delta q\xrightarrow{\Delta q\to0}\frac{1}{\Delta q}\int_{-\infty}^\infty e^{-\beta cq^2}dq=\frac{1}{\Delta q}\sqrt{\pi/\beta c} \Rightarrow \langle E\rangle=-Z^{-1}\partial Z/\partial\beta=\frac{1}{2}kT$

只适用经典系统且 $\Delta q\ll kT$. (7) Maxwell 分布 分子速率分布 $D(v)$, 概率 $(v<速<v+dv)=D(v)dv$. $D(v)\propto e^{-\frac{1}{2}mv^2\beta}\cdot\underbrace{4\pi v^2}_{对应v的\vec v个数}$

$1=\int_0^\infty D(v)dv=4\pi C\int_0^\infty v^2e^{-\frac{1}{2}mv^2\beta}dv=C(2\pi/m\beta)^{3/2} \Rightarrow C=(m\beta/2\pi)^{3/2}=(m/2\pi kT)^{3/2} \Rightarrow D(v)=(m/2\pi kT)^{3/2}\cdot4\pi v^2e^{-mv^2/2kT}$. $v_{max}=\sqrt{2kT/m}$, $\langle v\rangle=\sqrt{8kT/\pi m}$, $v_{rms}=\sqrt{3kT/m}$

(8) 配分函数. $F=-kT\ln Z$, $(\partial F/\partial T)_{V,N}=-S=\frac{F-U}{T} \Rightarrow S=-(\partial F/\partial T)_{V,N}$, $P=-(\partial F/\partial V)_{T,N}$, $\mu=(\partial F/\partial N)_{T,V}$ 对于可区分多粒子: $Z_{tot}=\prod_iZ_i$; 可区分 $Z_{tot}=\frac{1}{N!}\prod_iZ_i$

(9) 理想气体. N 个相同分子 $Z=\frac{1}{N!}Z_1^N$, 单个分子 $E=E_{tr}+E_{int} \Rightarrow Z_1=Z_{tr}Z_{int}$. 计算 Z_{tr}, 设分子是盒内波函数. 在边界为0 ⇒ 允许驻波 $\lambda_n=2L/n$, $n=1,2,\cdots \Rightarrow$

$P_n=h/\lambda_n=hn/2L \Rightarrow E_n=h^2n^2/8mL^2 \Rightarrow Z_{1D}=\sum_n\exp(-h^2n^2/8mL^2kT)\approx\int_0^\infty\exp(-h^2n^2/8mL^2kT)dn=\sqrt{2\pi mkT/h^2}\,L\equiv L/l_Q$, $l_Q\equiv h/\sqrt{2\pi mkT}$. 推至3D, $Z_{tr}=V/v_Q=V/l_Q^3$

$\therefore Z_1=V/v_Q\,Z_{int} \Rightarrow Z=\frac{1}{N!}(VZ_{int}/v_Q)^N \Rightarrow U=-\frac{\partial}{\partial\beta}\ln Z=-N\frac{\partial}{\partial\beta}\ln Z_{int}+N\frac{1}{v_Q}\frac{\partial v_Q}{\partial\beta}=U_{int}+\frac{3}{2}NkT$, $C_V=\partial U/\partial T=\partial U_{int}/\partial T+\frac{3}{2}Nk$; $F=-kT\ln Z=-NkT(\ln V-\ln N$

$-\ln v_Q+1)+F_{int}$; $P=-(\partial F/\partial V)_{N,T}=NkT/V$; $S=-(\partial F/\partial T)_{V,N}=Nk(\ln V/Nv_Q+\frac{5}{2})-\partial F_{int}/\partial T$; $\mu=(\partial F/\partial N)_{T,V}=-kT\ln V/Nv_Q-kT\ln Z_{int}$; $G=F+PV=N\mu$.

VII.量子统计 (1) Gibbs 因子. 允许交换能量+粒子 ($dN\neq0$) $\Rightarrow P(S_1)/P(S_2)=\Omega_R(S_1)/\Omega_R(S_2)=\exp\{[S_R(S_2)-S_R(S_1)]/k\}$. 原子相较热库很小, $S_R(S_2)-S_R(S_1)=dS_R$

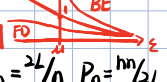
$\frac{1}{T}[dU_R+PdV_R-\mu dN_R]\approx\frac{1}{T}[E(S_2)-E(S_1)-\mu N(S_2)+\mu N(S_1)] \Rightarrow P(s)\propto\exp\{-[E(s)-\mu N(s)]/kT\} \Rightarrow \mathcal{Z}=\sum_se^{-\beta[E(s)-\mu N(s)]}$ (2) Boson, Fermion. 区分量子/经典

的条件 $Z_1\gg N \Leftrightarrow V/N\gg v_Q$. 即多个粒子占据相同单粒子态可能性忽略不计(经典). 量子条件下, Boson 可与同物质共享态(整数自旋), Fermion 不能(半整数自旋)

(3) Fermi-Dirac/Bose-Einstein 分布. 考虑一个由单粒子态组成的系统. 能量 ε 的态被 n 个粒子占据可能性 $P(n)=\mathcal{Z}^{-1}e^{-n(\varepsilon-\mu)\beta}$. 对于 Fermion, $n=0/1$, $\mathcal{Z}=1+e^{-(\varepsilon-\mu)\beta}$

$\Rightarrow \bar n_{FD}=0\cdot P(0)+1\cdot P(1)=1/(e^{\beta(\varepsilon-\mu)}+1)$; 对于 Boson, $n=0,1,2,\cdots$, $\mathcal{Z}=\sum_{n=0}^\infty e^{-n\beta(\varepsilon-\mu)}=1/(1-e^{-\beta(\varepsilon-\mu)}) \Rightarrow \bar n_{BE}=\sum_{n=0}^\infty n\cdot\mathcal{Z}^{-1}e^{-n\beta(\varepsilon-\mu)}$ $\underline{\underline{x=\beta(\varepsilon-\mu)}}$

$-\mathcal{Z}^{-1}\sum_{n=0}^\infty\frac{\partial}{\partial x}e^{-nx}=-\mathcal{Z}^{-1}\partial\mathcal{Z}/\partial x=-(1-e^{-x})\frac{\partial}{\partial x}(1-e^{-x})^{-1}=(e^x-1)^{-1}=1/(e^{\beta(\varepsilon-\mu)}-1)$; $\bar n_{BZ}=NP(s)=N/Z_1\,e^{-\varepsilon/kT}=e^{-\beta(\varepsilon-\mu)} \Rightarrow \beta(\varepsilon-\mu)\gg1$, 三种分布相同



(4) 简并 Fermi 气体. $T=0$ 时, FD 分布 ⇒ 阶跃, 仅所有低于 μ 的单粒子态被占. Fermi 气体简并, Fermi 能 $\varepsilon_F=\mu$. 假设自由粒子限制在 $V=L^3$ 内, $\lambda_n=2L/n$, $P_n=hn/2L$

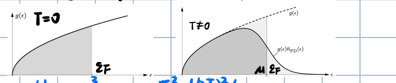
$\Rightarrow \varepsilon=h^2/8mL^2(n_x^2+n_y^2+n_z^2)$. (n_x,n_y,n_z) 能量空间近似 $\frac{1}{8}$ 球, 半径 $n_{max} \Rightarrow N=2\times\frac{1}{8}\times\frac{4}{3}\pi n_{max}^3$ (一个能量两个自旋) $\Rightarrow n_{max}=(3N/\pi)^{\frac{1}{3}} \Rightarrow \varepsilon_F=h^2n_{max}^2/8mL^2=h^2/8m(3N/\pi V)^{\frac{2}{3}}$

电子平均能量：$U=2\sum_{n_x}\sum_{n_y}\sum_{n_z}\varepsilon(\vec n)=2\int_0^{n_{max}}dn\int_0^{\pi/2}d\theta\int_0^{\pi/2}d\phi\, n^2\sin\theta\,\varepsilon(n)=\pi\int_0^{n_{max}}\varepsilon(n)n^2dn=\pi h^2/8mL^2\int_0^{n_{max}}n^4dn=\pi h^2 n_{max}^5/40mL^2=\frac35 N\varepsilon_F\Rightarrow \bar E=\frac35\varepsilon_F$

Fermi温度 $T_F=\varepsilon_F/k$，简并压 $P=-\partial U/\partial V=2N\varepsilon_F/5V=2U/3V>0$，来自Pauli不相容，对抗金属中的静电力. 压缩模量 $B=-V(\partial P/\partial V)_T=10U/9V$　$P(\varepsilon)=\frac12$的ε

态密度：对$U=\pi\int_0^{n_{max}}\varepsilon(n)n^2dn$换元：$\varepsilon=h^2/8mL^2\,n^2\Rightarrow n=\sqrt{8mL^2/h^2}\sqrt\varepsilon$，$dn=\sqrt{8mL^2/h^2}\cdot 1/2\sqrt\varepsilon\,d\varepsilon\Rightarrow T=0$时，$U=\int_0^{\varepsilon_F}\varepsilon[\frac\pi2(8mL^2/h^2)^{3/2}\sqrt\varepsilon]d\varepsilon=\int_0^{\varepsilon_F}\varepsilon g(\varepsilon)d\varepsilon$

$g(\varepsilon)\equiv\frac\pi2(8mL^2/h^2)^{3/2}\sqrt\varepsilon=3N/2\varepsilon_F^{3/2}\sqrt\varepsilon=g_0\sqrt\varepsilon$. $N=\int_0^{\varepsilon_F}g(\varepsilon)d\varepsilon$. 当T>0时，引入FD分布. $N=\int_0^\infty g(\varepsilon)\cdot 1/e^{\beta(\varepsilon-\mu)}+1\,d\varepsilon$，$U=\int_0^\infty\varepsilon g(\varepsilon)\cdot1/e^{\beta(\varepsilon-\mu)}+1\,d\varepsilon$. 此时$\mu<\varepsilon_F$



Sommerfield展开：$N=g_0\int_0^\infty\sqrt\varepsilon\,\bar n_{FD}(\varepsilon)d\varepsilon=\frac23g_0\varepsilon^{\frac32}\bar n_{FD}|_0^\infty+\frac23g_0\int_0^\infty\varepsilon^{\frac32}(-d\bar n_{FD}/d\varepsilon)d\varepsilon=\frac23g_0\int_0^\infty\beta\, e^x/(e^x+1)^2\varepsilon^{\frac32}d\varepsilon$
$=\frac23g_0\int_{-\mu/kT}^\infty e^x/(e^x+1)^2\varepsilon^{\frac32}dx$ ($x=\beta(\varepsilon-\mu)$)　$|\varepsilon-\mu|\gg kT$时作近似：把下限扩至$-\infty$，在$\varepsilon=\mu$处展开$\varepsilon^{\frac32}$:
$\varepsilon^{\frac32}=\mu^{\frac32}+\frac32(\varepsilon-\mu)\mu^{\frac12}+\cdots$. $N\approx\frac23g_0\int_{-\infty}^{+\infty}e^x/(e^x+1)^2[\mu^{\frac32}+\frac32xkT\mu^{\frac12}+\cdots]dx=\frac23g_0\mu^{\frac32}+\frac14g_0(kT)^2\mu^{-\frac12}\cdot\frac{\pi^2}{3}+\cdots=N(\mu/\varepsilon_F)^{\frac32}+N\cdot\frac{\pi^2}{8}(kT)^2/\varepsilon_F^{3/2}\mu^{1/2}+\cdots$

消去N，可知$\mu/\varepsilon_F\approx1$，修正项$\propto(kT/\varepsilon_F)^2$很小$\Rightarrow\mu$替换$\varepsilon_F$解出$\mu/\varepsilon_F=[1-\frac{\pi^2}{8}(kT/\varepsilon_F)^2+\cdots]^{2/3}=1-\frac{\pi^2}{12}(kT/\varepsilon_F)^2+\cdots$　同理$U=\frac35N\mu^{5/2}/\varepsilon_F^{3/2}+\frac{3\pi^2}{8}N(kT)^2/\varepsilon_F+\cdots$

代入μ：$U=\frac35N\varepsilon_F+\frac{\pi^2}{4}N(kT)^2/\varepsilon_F+\cdots$ (5)黑体辐射. 盒子中的电磁场可看作不同波长驻波相加. 每个波对应一个振子，允许能级$E_n=0,hf,2hf,\cdots$. 故单个振子配分$Z_1=1+e^{-\beta hf}+\cdots=1/1-e^{-\beta hf}\Rightarrow\langle E\rangle=-Z_1^{-1}\partial Z_1/\partial\beta=hf/e^{\beta hf}-1\Rightarrow$能量份数平均值$\bar n_{Pl}=1/e^{\beta hf}-1$ (Planck分布). 同时光子满足BE分布. 比较$\bar n_{BE}$，$\bar n_{Pl}$，$\varepsilon=hf\Rightarrow\mu=0$

类似电子气体，对光子的模式求和，注意一个波形对应2个光子：$\varepsilon=hcn/2L\Rightarrow N=2\sum_{n_x,n_y,n_z}1/e^{\beta hcn/2L}-1=\pi\int_0^\infty n^2/e^{\beta hcn/2L}-1\,dn=\pi(2LkT/hc)^3\int_0^\infty x^2/e^x-1\,dx$，$x=\beta hcn/2L$

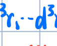
$U=2\sum_{n_x,n_y,n_z}\varepsilon\cdot\bar n_{Pl}=\frac\pi2\int_0^\infty hc/L\cdot n^3/e^{\beta hcn/2L}-1\,dn=V\int_0^\infty 8\pi/(hc)^3\,\varepsilon^3/e^{\beta\varepsilon}-1\,d\varepsilon$ ($\varepsilon=hcn/2L$，$V=L^3$) $\Rightarrow U/V=\int_0^\infty 8\pi/(hc)^3\,\varepsilon^3/e^{\beta\varepsilon}-1\,d\varepsilon$. 被积函数是单位光子能量的能量密度，称光子的谱 $u(\varepsilon)$　再换元 $x=\varepsilon/kT$. $U/V=8\pi(kT)^4/(hc)^3\int_0^\infty x^3/e^x-1\,dx=8\pi^5(kT)^4/15(hc)^3=aT^4\Rightarrow C_V=4aT^3$. $S(T)=\int_0^T C_V(T')/T'dT'=\frac43aT^3$，$P=-(\partial U/\partial V)_{S,N}=$
$\frac13(3S/4)^{4/3}a^{-1/3}V^{-4/3}=U/3V=\frac13aT^4$，$F=U-TS=-\frac13U$　热物体发射光子功率：考虑围绕孔的某个球壳，逃逸能量=球壳能量×逃逸概率$=\int_0^{2\pi}d\phi\int_0^{\pi/2}d\theta\cdot U/V$
$(Rd\theta)\cdot(R\sin\theta d\phi\cdot cdt)\times A\cos\theta/4\pi R^2=AU/4V\cdot cdt\Rightarrow$单位面积功率$=cU/4V=\sigma T^4$，$\sigma=2\pi^5k^4/15h^3c^2=5.67\times10^{-8}W\cdot m^{-2}\cdot K^{-4}$　也适用于由非反射表面辐射的光子

(6) Debye固体. 晶体中原子振动不独立，低温时类似声波，写出Planck分布$\bar n_{Pl}=1/e^{\beta\varepsilon}-1$，$\varepsilon=hc_sn/2L$. 对所有模式求和，这里上限不是∞，因为声波波长>原子间距
$\therefore n\le\sqrt[3]{N}$.　同样把正方体的n空间看作体积N的$\frac18$球. $n_{max}=(6N/\pi)^{\frac13}$. 则$U=3\sum_{n_x,n_y,n_z}\varepsilon\bar n_{Pl}$ (3来自每个$\vec n$ 3个极化态) $=\frac{3\pi}{2}\int_0^{n_{max}}hc_s/2L\cdot n^3/e^{\beta hc_sn/2L}-1\,dn$
令$x=\beta hc_sn/2L$，$x_{max}=\beta hc_sn_{max}/2LT=hc_s/2kT(6N/\pi V)^{\frac13}\equiv T_D/T$. 则$U=9NkT^4/T_D^3\int_0^{T_D/T}x^3/e^x-1\,dx$.　$T\gg T_D$时，$U\approx9NkT^4/T_D^3\int_0^{T_D/T}x^2dx=3NkT\Rightarrow C_V=3Nk$. 低温$T\ll T_D$时，
$U\approx9NkT^4/T_D^3\int_0^\infty x^3/e^x-1\,dx=3\pi^4Nk/5T_D^3\cdot T^4\Rightarrow C_V=12\pi^4Nk/5T_D^3\,T^3\propto T^3$. 中间温度$C_V=9Nk(T/T_D)^3\int_0^{T_D/T}x^4e^x/(e^x-1)^2dx$　对比Einstein固体$C_V=3Nk(\beta\varepsilon)^2e^{\beta\varepsilon}/(e^{\beta\varepsilon}-1)^2$

(7) BE凝聚. 当$T<T_c$时，激发态能容纳的粒子↓，基态出现宏观占据$N_0\sim O(N)$. 回忆态密度$g(\varepsilon)=\frac\pi4(8mL^2/h^2)^{3/2}\sqrt\varepsilon$ (一种自旋)，$\bar n_{BE}=1/e^{\beta(\varepsilon-\mu)}-1$. 则
$N_{ex}=\sum_{\varepsilon>0}\bar n(\varepsilon)=\int_0^\infty g(\varepsilon)\cdot1/e^{\beta(\varepsilon-\mu)}-1\,d\varepsilon$. $\mu\to0^-$时(这时激发态容量最大)，$N_{ex}^{max}(T)=g_0\int_0^\infty\sqrt\varepsilon/e^{\beta\varepsilon}-1\,d\varepsilon\propto\int_0^\infty\sqrt u/e^u-1\,du=\int_0^\infty\sqrt u\sum_{n=1}^\infty e^{-nu}du=\sum_{n=1}^\infty\int_0^\infty\sqrt u\, e^{-nu}du$
$=\sum_{n=1}^\infty\Gamma(\frac32)/n^{3/2}=\Gamma(\frac32)\zeta(\frac32)\Rightarrow N_{ex}^{max}=V/\lambda^3\,\zeta(\frac32)$ 收敛. 于是$N_{ex}^{max}(T)\propto T^{3/2}$，T足够小时，$N_{ex}^{max}<N$，大部分粒子进入基态，发生凝聚. 这个T就是$T_c$
$T_c=[N/V\zeta(3/2)]^{2/3}h^2/2\pi mk$ (此时$N_0\to0$，$N_{ex}^{max}=N$).　$T<T_c$时，$N_{ex}=N(T/T_c)^{3/2}$，$N_0=N[1-(T/T_c)^{3/2}]$　二维时无BEC：回忆态密度定义$g(\varepsilon)d\varepsilon=[\varepsilon,\varepsilon+d\varepsilon]$
d维n-空间中，半径n的球壳表面积$\propto n^{d-1}dn$，而$\varepsilon\propto n^2\Rightarrow dn/d\varepsilon\propto1/\sqrt\varepsilon\Rightarrow g(\varepsilon)\propto n^{d-1}dn/d\varepsilon\propto\varepsilon^{d/2-1}$　而$N_{ex}^{max}(T)\propto\int_0^\infty\varepsilon^{d/2-1}/e^{\beta\varepsilon}-1\,d\varepsilon\propto\int_0^\infty x^{d/2-1}/e^x-1\,dx=\Gamma(\frac d2)\zeta(\frac d2)$
因此d>2时$\zeta(\frac d2)$收敛，有BEC，$d\le2$时$\zeta(\frac d2)$发散，无BEC. 凝聚$\Leftarrow g(\varepsilon)\to0,\varepsilon\to0$

VIII. 相互作用的粒子　(1)维里展开：$PV=nRT(1+B/V/n+C/(V/n)^2+\cdots)$. $B=-2\pi\int_0^\infty r^2[e^{-\beta u(r)}-1]dr$，$u(r)=u_0[(r_0/r)^{12}-2(r_0/r)^6]$. 用van der Waals估计B,C：$(P+an^2/V^2)(V-nb)=nRT\Rightarrow PV=nRT(1/1-nb/V-an/RTV)\approx nRT(1+nb/V+n^2b^2/V^2-an/RTV)\Rightarrow B(T)=b-a/RT$，$C(T)=b^2$

(2) Ising模型　N个偶极子，s_i为态(+1/-1)，只考虑相邻作用$-s_i\cdot s_j\cdot\varepsilon\Rightarrow U=-\varepsilon\sum_{adj}s_i\cdot s_j$，$Z=\sum_{\{s_i\}}e^{-\beta U}$.　1维下$U=-\varepsilon(s_1s_2+s_2s_3+\cdots+s_{N-1}s_N)$
$\Rightarrow Z=\sum_{s_1}\cdots\sum_{s_N}e^{\beta\varepsilon s_1s_2}\cdots e^{\beta\varepsilon s_{N-1}s_N}=\sum_{s_1}\cdots\sum_{s_{N-1}}e^{\beta\varepsilon s_1s_2}\cdots e^{\beta\varepsilon s_{N-2}s_{N-1}}\cdot2\cosh(\beta\varepsilon)=\sum_{s_1}1\cdot(2\cosh\beta\varepsilon)^{N-1}=2^N(\cosh\beta\varepsilon)^{N-1}\approx[2\cosh(\beta\varepsilon)]^N\Rightarrow$
$\bar U=-\partial/\partial\beta\ln Z=-N\varepsilon\tanh\beta\varepsilon$. 与双状态顺磁体一致.　对2维，引入平均场近似：$\bar s=\tanh(\beta\varepsilon n\bar s)$ 受周围n个影响. $\beta\varepsilon n<1\Leftrightarrow kT>n\varepsilon$
有稳定解$\bar s=0$；$\beta\varepsilon n>1\Leftrightarrow kT<n\varepsilon$，有3解，$\bar s=0$不稳定$\Rightarrow$自发磁化. $kT_c=n\varepsilon$.　T_c附近$M\propto(T_c-T)^{\frac12}$ ($\tanh x=x-\frac13x^3$)；$\chi=(\partial M/\partial B)_T$. 外加B，则
$\bar s=\tanh(\beta\varepsilon n\bar s+\beta\mu_BB)$，展开$\Rightarrow\chi\propto(T-T_c)^{-1}$

IX. 补充：(1)对应系数. 由热力学恒等式推出：$(\partial U/\partial S)_{V,N}=T$，$(\partial U/\partial V)_{N,S}=-P$，$(\partial U/\partial N)_{S,V}=\mu$；$(\partial H/\partial S)_{P,N}=T$，$(\partial H/\partial P)_{N,S}=V$，$(\partial H/\partial N)_{S,P}=\mu$；$(\partial F/\partial T)_{V,N}=-S$
$(\partial F/\partial V)_{N,T}=-P$，$(\partial F/\partial N)_{T,V}=\mu$；$(\partial G/\partial T)_{P,N}=-S$，$(\partial G/\partial P)_{N,T}=V$，$(\partial G/\partial N)_{T,P}=\mu$. (2) Maxwell关系式：$U\Rightarrow(\partial T/\partial V)_S=-(\partial P/\partial S)_V$，$H\Rightarrow(\partial T/\partial P)_S=(\partial V/\partial S)_P$
$F\Rightarrow(\partial S/\partial V)_T=(\partial P/\partial T)_V$，$G\Rightarrow(\partial S/\partial P)_T=-(\partial V/\partial T)_P$　(3)技巧：$S=S(T,V)\Rightarrow dS=(\partial S/\partial T)_VdT+(\partial S/\partial V)_TdV$，沿等P过程，$(\partial S/\partial T)_P=(\partial S/\partial T)_V+(\partial S/\partial V)_T(\partial V/\partial T)_P$
$\Rightarrow C_P=C_V+T(\partial S/\partial V)_T(\partial V/\partial T)_P$. 再代入Maxwell：$(\partial S/\partial V)_T=(\partial P/\partial T)_V$ 以及轮换：$(\partial P/\partial T)_V=-(\partial V/\partial T)_P/(\partial V/\partial P)_T\Rightarrow C_P-C_V=-T(\partial V/\partial T)_P^2/(\partial V/\partial P)_T$　(4)热力学方程：
设$U=U(T,V)$，$dU=(\partial U/\partial T)_VdT+(\partial U/\partial V)_TdV=C_VdT+[T(\partial P/\partial T)_V-P]dV$ (利用$dU=TdS-PdV$及$(\partial S/\partial V)_T=(\partial P/\partial T)_V$). 同理$dH=C_PdT+[-T(\partial V/\partial T)_P+V]dP$
(5)理想气体性质：$(\partial U/\partial V)_T=T(\partial P/\partial T)_V-P=0\Rightarrow U=U(T)$. 同理$H=H(T)$. 另外$(\partial C_P/\partial P)_T=[\partial/\partial P(\partial H/\partial T)_P]_T=[\partial/\partial T(\partial H/\partial P)_T]_P=\{\partial/\partial T[V-T(\partial V/\partial T)_P]\}_P=-T(\partial^2V/\partial T^2)_P=0$
同理$C_V=T(\partial^2P/\partial T^2)_V=0\Rightarrow C=C(T)$　(6)二维气体　$\Omega=1/N!\cdot A^N/h^{2N}\cdot2\pi^N/(N-1)!\,(2mU)^{N-\frac12}\approx1/N!\cdot A^N/h^{2N}\cdot\pi^N/N!\,(2mU)^N$.　$S=Nk[\ln(A/N\cdot2\pi mU/Nh^2)+2]$. $T=U/Nk\Rightarrow f=2$
$\mu=-kT\ln[A/N\cdot2\pi mkT/h^2]$　(7) Einstein固体：不作$q\ll N$(或≫)近似，$\Omega(N,q)\approx(q+N/q)^{q+N}/q^qN^N\Rightarrow S\Rightarrow T$ (这里$U=q\varepsilon$) $\Rightarrow U=N\varepsilon/e^{\varepsilon/kT}-1\Rightarrow C=N\varepsilon^2/kT^2\,e^{\varepsilon/kT}/(e^{\varepsilon/kT}-1)^2$
$q\ll N$时，$C_V=Nk(\varepsilon/kT)^2e^{-\varepsilon/kT}$ (比较Debye模型)　(8) Saha方程：$H\leftrightarrow p+e$　视为理想气体，$\mu=-kT\ln[kT/P(2\pi mkT/h^2)^{3/2}]$，但$\mu_H$多减去电离能I. $kT\ln(P_HP^\circ/P_pP_e)=\Delta\mu$
$\Rightarrow P_p/P_H=kT/P_e(2\pi m_ekT/h^2)^{3/2}e^{-I/kT}$　(9)二维气体的Maxwell分布：$D(v)=C\cdot2\pi v\cdot e^{-mv^2/2kT}$，$1=\int_0^\infty D(v)v=2\pi kTC/m\Rightarrow C=m/2\pi kT\Rightarrow D(v)=m/kT\cdot v\cdot e^{-mv^2/2kT}$，
$v_{max}=\sqrt{kT/m}$ (注意最可能的$\vec v$是0)，$\langle v\rangle=\int_0^\infty vD(v)dv=\sqrt{\pi kT/2m}$，$v_{rms}=\sqrt{2kT/m}$　(10)正则系综三级中心矩：$\langle E^n\rangle=Z^{-1}\sum_sE^n(s)e^{-\beta E(s)}=(-1)^nZ^{-1}\partial^nZ/\partial\beta^n$.
$\langle(E-\langle E\rangle)^3\rangle=\langle E^3\rangle-3\langle E^2\rangle\langle E\rangle+2\langle E\rangle^3=-Z^{-1}Z'''+3Z^{-2}Z''\cdot Z'-2Z^{-3}(Z')^3=-\partial^3(\ln Z)/\partial\beta^3$　回忆$\sigma_E^2=\partial^2(\ln Z)/\partial\beta^2=kT^2C_V$，则$\langle(\Delta E)^3\rangle=-\partial(kT^2C_V)/\partial\beta=-\partial T/\partial\beta\,\partial(kT^2C_V)/\partial T$
$=k^2[T^4(\partial C_V/\partial T)_V+2T^3C_V]$.　对于理想气体，$\langle(E-\langle E\rangle)^3\rangle/\langle E\rangle^3=8/N^2 9$　(11)巨正则系综N的期望. $\langle N\rangle=kT/\mathcal Z\,\partial\mathcal Z/\partial\mu$，$\langle N^2\rangle=(kT)^2/\mathcal Z\,\partial^2\mathcal Z/\partial\mu^2\Rightarrow\sigma_N^2=kT\,\partial\langle N\rangle/\partial\mu$
对于理想气体，$\sigma_N=\sqrt N\Rightarrow$相对涨落$1/\sqrt N$. (12)巨自由能. $\Phi\equiv U-TS-\mu N=-PV$. 对于热+扩散平衡系，$\Phi$最小化. $d\Phi=-SdT-PdV-Nd\mu$. $(\partial\Phi/\partial T)_{V,\mu}=-S$，$(\partial\Phi/\partial V)_{T,\mu}=-P$
$(\partial\Phi/\partial\mu)_{T,V}=-N$. 考虑$\tilde\Phi=-kT\ln\mathcal Z$，$(\partial\tilde\Phi/\partial\mu)_{T,V}=-kT\mathcal Z^{-1}\partial\mathcal Z/\partial\mu=-\langle N\rangle\Rightarrow\Phi=\tilde\Phi=-kT\ln\mathcal Z$　(13)三维盒中Fermi气体，受到B磁场，态密度$g(\varepsilon)=\frac{g_0}{2}(\sqrt{\varepsilon+\delta}+\sqrt{\varepsilon-\delta})$
$g_0=3N/2\varepsilon_F^{3/2}$，$\delta=\mu_BB\Rightarrow M=\mu_B(N_\uparrow-N_\downarrow)\approx\mu_B\cdot2\cdot\delta\cdot\frac12g_0\sqrt{\varepsilon_F}=3N\mu_B^2B/2\varepsilon_F$.　$T\ll T_F$时，$M=M_0[1-\frac{\pi^2}{12}(kT/\varepsilon_F)^2]$ (Sommerfield)　(14)二维Fermi气体，限制在$A=L^2$中. $\varepsilon_F=h^2n_{max}^2/8mA$. $N=2\cdot\pi n_{max}^2/4\Rightarrow\varepsilon_F=h^2N/4\pi mA$. $U=\pi h^2n_{max}^4/32mA=\frac12N\varepsilon_F$. $g(\varepsilon)=4\pi mA/h^2=N/\varepsilon_F$；$kT\ll\varepsilon_F$，T↗，$\mu\approx\varepsilon_F$；$kT\gg\varepsilon_F$，T↗，μ↘，与理想气体相同
$N=\int_0^\infty g(\varepsilon)\bar n_{FD}d\varepsilon=g(\varepsilon)kT\ln(1+e^{\mu/kT})\Rightarrow\mu=kT\ln(e^{\varepsilon_F/kT}-1)$；T>0时，$U=N/\varepsilon_F\int_0^\infty\varepsilon\bar n_{FD}d\varepsilon\approx\frac12N\varepsilon_F+\frac{\pi^2}6N(kT)^2/\varepsilon_F\Rightarrow C_V=\pi^2Nk^2T/3\varepsilon_F$　(15)光子气体补充. 态密度
$g(\varepsilon)=8\pi V\varepsilon^2/(hc)^3$，总数$N=2\sum_{n_x,n_y,n_z}\bar n_{Pl}=2\cdot\frac\pi2\cdot\int_0^\infty n^2/e^{\beta hcn/2L}-1\,dn=8\pi V(kT/hc)^3\int_0^\infty x^2/e^x-1\,dx$.　(16)二维固体表面：1种极化. $U=\int_0^{\sqrt N}dn_x\int_0^{\sqrt N}dn_y\,\varepsilon/e^{\beta\varepsilon}-1=\frac\pi2\int_0^{n_{max}}n\varepsilon/e^{\beta\varepsilon}-1\,d\varepsilon$
$n_{max}=\sqrt{4N/\pi}$ . 令$x=\varepsilon/kT=hc_sn/2LkT$，$x_{max}=hc_s/2LkT\sqrt{4N/\pi}\equiv T_D/T$. $U=2NkT^3/T_D^2\int_0^{T_D/T}x^2/e^x-1\,dx$. $T\gg T_D$，$U\approx NkT$；$C_A=2NkT^2/T_D^2\int_0^{x_m}x^3e^x/(e^x-1)^2dx$　(17)集团展开.
用位置+动量表示分子态. $Z=1/N!\,1/h^{3N}\int d^3r_1\cdots d^3r_Nd^3p_1\cdots d^3p_Ne^{-\beta U}=Z_{ideal}\cdot1/V^N\int d^3r_1\cdots d^3r_Ne^{-\beta U_{pot}}=Z_{ideal}\cdot Z_c$. 令$e^{-\beta u_{i,j}}=1+f_{i,j}$，$Z_c=V^{-N}\int d^3r_1\cdots d^3r_N\prod(1+f_{i,j})$
$=V^{-N}\int d^3r_1\cdots d^3r_N(1+\sum f_{i,j}+\sum f_{i,j}f_{k,l})=1+\mathrm{I}+\Lambda+(\mathrm{I}\,\mathrm{I})+\triangle+\cdots$.　$\mathrm{I}=\frac12N(N-1)/V^2\int d^3r_1d^3r_2f_{1,2}$；第i个点→$N+1-i/V\int d^3r_i$；连i,j点线⇒$f_{i,j}$；除对称因子(编号数)
$\Lambda=\frac1{2V^3}N(N-1)(N-2)\int d^3r_1d^3r_2d^3r_3f_{1,2}f_{2,3}$　$(\mathrm{I}\,\mathrm{I})=\frac18N(N-1)(N-2)(N-3)/V^4\int d^3r_1\cdots d^3r_4f_{1,2}f_{3,4}$. 对于低密度，$1+\mathrm{I}+(\mathrm{I}\,\mathrm{I})+\cdots\approx1+\mathrm{I}+\frac1{2!}(\mathrm{I})^2+\cdots=\exp(\mathrm{I})\Rightarrow Z_c=\exp(1+\mathrm{I}+\triangle+\square+\cdots)$

X. (1) $\int_{-\infty}^\infty e^{-x^2}dx=\sqrt\pi$. $\int_0^\infty e^{-ax^2}dx=\frac12\sqrt{\pi/a}$. $\int_0^\infty x^2e^{-ax^2}dx=\frac14\sqrt{\pi/a^3}$ (Feymann积分法). $\int_0^\infty xe^{-ax^2}dx=\frac1{2a}$，$\int_0^\infty x^ne^{-ax^2}dx=\frac12a^{-n+1/2}\Gamma(n+1/2)$　(2) $\Gamma(n+1)=\int_0^\infty x^ne^{-x}dx$
(3)d维超球：$A_d(r)=2\pi^{d/2}/\Gamma(d/2)\,r^{d-1}$，$V_d(r)=\pi^{d/2}/\Gamma(d/2+1)\,r^d$　(4)量子统计积分. $\int_0^\infty x^n/e^x-1\,dx=\sum_{k=1}^\infty\int_0^\infty x^ne^{-kx}dx=\Gamma(n+1)\zeta(n+1)$，$\int_0^\infty x^n/e^x+1\,dx=(1-2^{-n})\Gamma(n+1)\zeta(n+1)$. $\zeta(2)=\pi^2/6$
$\zeta(4)=\pi^4/90$；算Fermi气体时，$\int_{-\infty}^\infty x^2e^x/(e^x+1)^2dx=2[0+\int_0^\infty2x(e^x+1)^{-1}dx]=\pi^2/3$　(6) Taylor展开：$e^x=\sum_{n=0}^\infty x^n/n!$. $\ln(1-x)=-\sum_{n=1}^\infty x^n/n$　$1/1-x=\sum_{n=0}^\infty x^n$，$1/(1-x)^2=\sum_{n=1}^\infty nx^{n-1}$，
$1/(1-x)^3=\sum_{n=2}^\infty\frac12n(n-1)x^{n-2}$，$(1+x)^{\frac12}=1+\frac12x-\frac18x^2+\frac1{16}x^3+\cdots$，$(1+x)^{-\frac12}=1-\frac12x+\frac38x^2-\frac5{16}x^3+\cdots$，$\sin x=\sum_{n=0}^\infty\frac{(-1)^n}{(2n+1)!}x^{2n+1}$，$\cos x=\sum_{n=0}^\infty\frac{(-1)^n}{(2n)!}x^{2n}$，$\tan x=x+x^3/3+2x^5/15+\cdots$. 双曲正"-"